

Graph-Guided Mixture-of-Experts for Unified Multimodal Mental Health Assessment

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Abstract

We propose a unified multimodal architecture for mental health assessment that jointly learns from depression, sentiment, emotion, and personality across three public benchmarks: DAIC-WOZ (clinical interviews, $n = 189$), CMU-MOSEI (YouTube videos, $n = 22,777$), and ChaLearn First Impressions (video clips, $n = 10,000$). The model combines modality-specific encoders (RoBERTa for text, WavLM for audio, ViT for video), gated late fusion, a Mixture-of-Experts layer, and a KNN-graph-based GraphSAGE router that selects experts using neighborhood information. Graph-based routing has a *task-dependent* effect against a non-graph MMoEEx baseline that we verified is a real, working classifier on every task (an earlier version of this baseline was silently capped at chance-level DAIC performance, AUROC 0.493, by an unused temperature-balancing weight in the training data loader—a bug we diagnose and fix as part of this work; the corrected baseline reaches DAIC AUROC 0.573, consistent with a 0.671 AUROC unimodal-text-only ablation). Across five leakage-safe graph configurations, graph routing *underperforms* this working baseline on DAIC depression AUROC (0.471–0.551 vs. 0.573) and FI personality CCC (0.478–0.531 vs. 0.571 without), while showing a small, consistent *gain* on MOSEI sentiment CCC (0.486–0.512 vs. 0.493) and emotion AUROC (0.713–0.742 vs. 0.703). A graph-homophily diagnostic explains the split: the routing graph carries no

more DAIC-label signal than random pairing, so no router could help there; it carries substantial signal for MOSEI and FI, which the router converts into a real gain for MOSEI but still fails to exploit for FI; a learned per-task routing weight narrows but does not close the DAIC gap, and a $K \in \{5, 10, 15, 20\}$ sensitivity sweep shows no configuration recovers a DAIC benefit. We report this as a rigorously investigated, task-dependent negative result. Separately, cross-attention fusion does not reproduce a reported gain over gated fusion under our evaluation protocol, and LLM-based encoders (Mistral-7B, CLAP, LLaVA) show a task-dependent pattern rather than a uniform win: substantial point-estimate gains on MOSEI sentiment and emotion recognition, a DAIC gain only from the full LLM stack (partial swaps underperform classical), and a uniform loss on FI personality across every LLM level—none of the DAIC comparisons reach statistical significance at this sample size. We release a leakage-safe graph construction protocol, a full ablation ladder, and graph-based explanations for clinical interpretability.

1 Introduction

Mental health assessment from multimodal signals is a cornerstone challenge in affective computing and clinical informatics. Depression, the primary focus of this work, affects over 280 million people worldwide [1] and is routinely assessed through clinical interviews that combine speech, facial expres-

sions, and verbal content. Meanwhile, related affective signals—sentiment, emotion, and apparent personality—provide complementary supervision signals that can enrich mental health models through multitask learning [2].

Our central hypothesis has two parts, which we evaluate separately rather than conflating. First, we hypothesize that mental health assessment can be made more *transparent* by characterizing a person’s affective and psychological profile—sentiment, emotion, and apparent personality—jointly with a depression assessment, rather than reporting a single opaque depression score in isolation: the auxiliary constructs give a clinician additional, more legible context (e.g., is a low mood reading accompanied by high negative sentiment and low extraversion, or is it isolated?) for interpreting the primary prediction. Second, we hypothesize that a graph over similar individuals can provide both a *visual* (subgraph diagrams, per-modality attribution plots) and a *machine-readable* (structured case records: SHAP values, neighbor identities and distances, routing weights) account of which factors and which comparable cases informed a given prediction, complementing the auxiliary-construct characterization with instance-level explanation. This is a claim about transparency, evaluated separately from the further hypothesis that graph-based *routing* improves raw predictive performance (Section 6)—a model can be well-explained without being more accurate, and our results show these are empirically distinct questions in this setting: the performance hypothesis is not supported, while the transparency mechanisms (multi-construct characterization, graph-based visual and machine-readable explanation) are implemented and evaluated for faithfulness in Section 8. We do not claim that apparent personality and clinical depression measure the same construct—personality is auxiliary context, not a depression proxy. The proposed system is intended as a decision-support tool and should not replace clinical diagnosis; all results require prospective validation before clinical deployment.

Figure 1 details the software architecture of our experimental pipeline, moving from data extraction through unimodal baselines into the joint multitask

training implementation. The central empirical question of our pipeline is the graph-routing investigation (Section 6). To isolate its effect, we compare five graph-routing ablation variants (V0–V4) against non-graph MMoEEx and KNN-voting baselines. When initial results showed no improvement on DAIC/MOSEI and degradation on FI personality, we deployed a homophily diagnostic comparing real graph edges against random pairing, alongside targeted fixes including a K -sensitivity sweep ($K = 5, 10, 15, 20$) and learning a per-task λ weight instead of fixing it at 0.5. These diagnostics confirmed that graph routing does not help in this setting because the required signal is either absent or unused. Downstream LLM, calibration, domain-adaptation, and explainability analyses then follow.

This work makes three contributions:

1. **A rigorously investigated negative result for graph-gated MoE routing:** across five leakage-safe graph configurations, a K -sensitivity sweep, and a learned-routing-weight variant, graph-based expert routing does not improve over a non-graph MMoEEx baseline, and degrades FI personality prediction specifically. We support this with a graph-homophily diagnostic that localizes *why* (Section 6), rather than reporting a single negative number without investigation.
2. **Leakage-safe graph construction protocol:** Three graph construction modes (inductive, split-local, transductive) under strict subject-independent splits, with formal validation that prevents test-label leakage through graph edges.
3. **Comprehensive empirical analysis:** A systematic ablation ladder spanning unimodal baselines, three fusion strategies, MoE variants, five graph configurations, and LLM encoder replacements, including a comparison showing LLM-based encoders outperform both classical encoders and graph routing on every task.

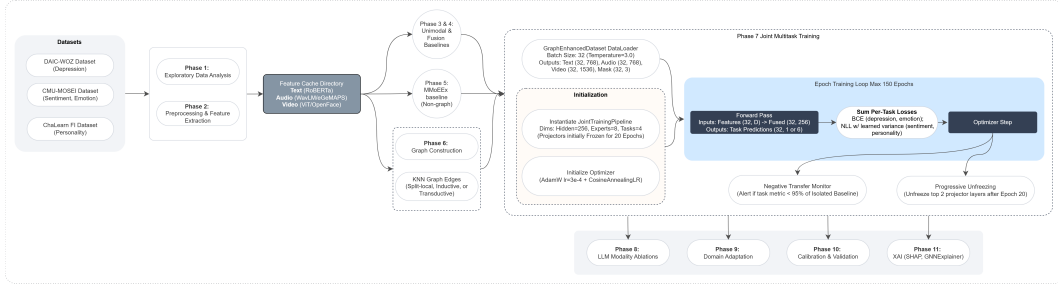


Figure 1: Implementation details of the experimental pipeline and joint multitask training. The diagram shows the progression from dataset caching through the non-graph MMoEEx baseline into the Phase 7 multitask epoch loop—including the negative-transfer monitor and progressive unfreezing mechanisms—and concluding with downstream domain adaptation, calibration, and XAI validation layers.

2 Related Work

Multimodal Fusion. Early fusion for mental health used simple feature concatenation. Later approaches adopted late fusion with learned modality gates or Low-Rank Multimodal Fusion (LMF) [3]. Cross-modal attention mechanisms have also been proposed for depression detection, but in our experimental configuration, we do not observe a benefit from cross-attention over gated fusion (Section 6).

Mixture of Experts. The sparsely-gated MoE layer [4] routes each input to a subset of experts. Multitask MoE (MMoE) extends this with task-specific gates [5], and MMoEEx adds expert exclusivity and orthogonality regularization [6]. We build on MMoEEx but add graph-based routing.

Graph Neural Networks for Routing. GraphSAGE [7] performs inductive representation learning on graphs. We use GraphSAGE as a router: aggregating features from K nearest neighbors to produce a graph context vector that modulates expert weights. Wang et al. [8] similarly combine GNN-based experts with explicit diversity modeling, though as GNN experts rather than a graph-conditioned router over standard MLP experts as here. Whether a KNN routing graph is useful depends on its label homophily/informativeness [9]; when it is not, graph rewiring or learned graph structure [10] are natural

remedies we discuss in Section 9. Task-affinity-aware grouping [11] and expert-choice routing [12] are two further candidate fixes we did not yet implement.

LLM-Enhanced Encoders. Large language models have been applied to depression detection via instruction-tuned text encoders and multi-modal audio-language models [13]. We evaluate LLM-enhanced encoders in our ablation track (levels L1–L5).

Multi-Construct Characterization and Graph-Based Interpretability. Interpretability for mental health assessment has been pursued by jointly modeling depression alongside related affective constructs, and by grounding graph structure in psychologically meaningful units: Vyalla et al. [14] explicitly distinguish personality trait context from acute depression symptoms via a psychologically-grounded, interpretable graph model, closely related to our multi-task characterization hypothesis (Section 1). For explaining LLM-based reasoning specifically, chain-of-thought and related structured-reasoning techniques have been shown to improve mental-health text classification [15], but can also be unfaithful or reduce reliability when reasoning is inserted into the clinical prediction pipeline itself [16] a distinction we return to in Section 8, where LLM narratives are used purely as *post-hoc* explanation of an independently-computed prediction, not as part

of the prediction process.

Depression Detection on DAIC-WOZ. The DAIC-WOZ corpus [17] contains 189 clinical interviews with PHQ-8 labels. Prior work spans graph-based text models [18], multiple-instance learning over clinical interviews [19], and hierarchical graph-attention fusion of text and audio [20]. Reported metrics vary widely by evaluation protocol (F1 vs. AUROC, inclusion of therapist/interviewer prompts, differing label thresholds), so we do not attempt a leaderboard-style numeric comparison and instead report AUROC under our own fixed, leakage-safe protocol throughout.

3 Datasets

DAIC-WOZ contains 189 sessions (107 train, 35 val, 47 test) of human-agent interviews for depression assessment. Each session has a PHQ-8 binary label (depression ≥ 10). We use subject-independent splits: no participant appears in more than one split. Features: RoBERTa-base text (768-dim), WavLM-large audio (1536-dim), eGeMAPS (88-dim), ViT-B/16 video (1536-dim), OpenFace AUs (70-dim).

CMU-MOSEI [21] contains 22,777 utterance-level samples (16,265 train, 1,869 val, 4,643 test). Labels: continuous sentiment and multi-label emotion (happiness, sadness, anger, fear, disgust, surprise). MOSEI is 120 $\times$ larger than DAIC, mitigated by temperature-balanced sampling (T=3.0).

ChaLearn First Impressions [22] contains 10,000 video clips (6,000 train, 2,000 val, 2,000 test) with Big-Five apparent personality labels. Personality is not the same construct as clinical depression; FI serves as auxiliary supervision.

Leakage Protocol. All splits are subject-independent. The graph is constructed separately for each split (split-local mode) or only on training data (inductive mode) to prevent test-label leakage through graph edges.

4 Architecture

Table 1: Notation.

Symbol	Meaning
K	Nearest neighbors (graph)
M	Number of experts (8)
d	Embedding dimension (256)
r	Low-rank bottleneck (8–16)
λ	Graph routing weight
σ_t	Task-specific uncertainty
G	KNN graph
r_i	Graph routing vector

Figure 2 gives an overview of the full pipeline.

Modality Encoders. Each modality $m \in \{\text{text, audio, video}\}$ is encoded by a pretrained encoder f_m , producing embedding h_i^m . A projection layer P_m maps to a common $d = 256$ -dimensional space:

$$h_i^{m,\text{proj}} = P_m(f_m(x_i^m)) \in \mathbb{R}^{256} \quad (1)$$

Gated Late Fusion. Given projected modality embeddings, we apply a modality mask $m_i \in \{0, 1\}^3$ and compute learned gate weights:

$$\alpha_i = \text{softmax}(W_\alpha \cdot \text{concat}(h_i^{\text{text}}, h_i^{\text{audio}}, h_i^{\text{video}}) + b_\alpha) \quad (2)$$

$$h_i^{\text{fused}} = \sum_m \frac{\alpha_i \odot m_i}{\|\alpha_i \odot m_i\|_1 + \epsilon} [m] \cdot h_i^{m,\text{proj}} \quad (3)$$

Each dataset is evaluated under a fixed, dataset-native routing policy chosen from our fusion ablation (Section 6): DAIC uses text-only routing ($h_i^{\text{fused}} = h_i^{\text{text}}$), FI uses video-only routing, and MOSEI uses the full gated fusion above. This is a deliberate architectural choice, not a missing-data fallback: text alone is DAIC’s strongest single modality and video alone is FI’s, so routing each dataset through its best-performing modality avoids diluting the fused representation with weaker signal on datasets too small to learn a reliable gate over all three. A direct consequence, verified in Section 8, is that a modality outside a dataset’s native routing contributes *exactly*

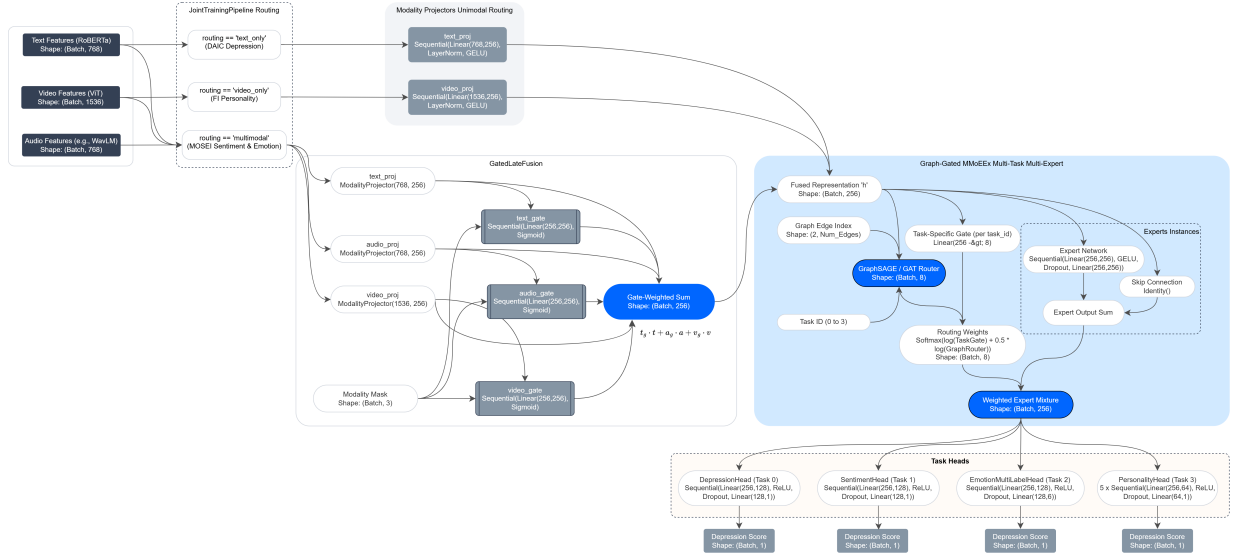


Figure 2: Unified multimodal graph-gated MoE architecture. Modality-specific encoders (RoBERTa/WavLM/ViT) project to a common $d = 256$ space, routed per dataset (text-only for DAIC, video-only for FI, gated late fusion for MOSEI) into h_i^{fused} ; the MMoEEx expert bank (8 experts, hidden dimension 256) and a KNN-graph GraphSAGE/GAT router jointly determine expert routing weights in log-space ($\lambda = 0.5$); task heads predict depression, sentiment, emotion, and personality.

zero to that dataset’s predictions and to any attribution computed on them (audio and video for DAIC; text and audio for FI) not merely a small or negligible contribution.

MMoEEx Expert Bank. We instantiate $M = 8$ expert MLPs $\{E_1, \dots, E_8\}$ with residual connections, hidden dimension 256 throughout. Two routing configurations are used across our experiments, differing in gating rather than expert size. In the non-graph MMoEEx baseline (Section 6, “MMoEEx vs. Baselines”), experts are hard-isolated per task group: depression uses $\{E_1, E_2\}$, MOSEI sentiment and emotion share $\{E_3, E_4\}$, and personality uses $\{E_5, E_6\}$; E_7, E_8 are allocated but unused under this isolation mask. In the graph-routed variants (V0–V4, our primary reported configuration), we disable hard isolation: task-specific gates $g_t : \mathbb{R}^{256} \rightarrow \mathbb{R}^8$ produce a full softmax over all 8 experts, combined with the graph router in log-space (Section 4).

KNN Graph Construction. We construct an undirected KNN graph $G = (V, E)$ over all training samples using cosine similarity. Three construction modes are tested:

- **Inductive (V0, V3):** Graph on training only. Test nodes connect to K nearest training neighbors.
- **Split-local (V1, V4):** Separate graphs per split. No cross-split edges.
- **Transductive (V2):** Graph on all splits. For ablation only.

GraphSAGE Router. Two-layer neighborhood aggregation produces a graph routing vector:

$$h_i^{(1)} = \sigma(W^{(1)} \cdot \text{Mean}(\{h_j^{\text{fused}} : j \in \mathcal{N}(i)\} \cup \{h_i^{\text{fused}}\})) \quad (4)$$

$$r_i = \text{softmax}(W_r h_i^{(2)} + b_r) \in \mathbb{R}^M \quad (5)$$

The final expert weights combine task gate and graph router in log-space:

$$\tilde{w}_{t,i} = \text{softmax}(\log g_t(h_i) + \lambda \cdot \log r_i) \quad (6)$$

λ is fixed to 0.5 (not tuned) in all reported graph-routed experiments.

Task Heads. Depression: binary classifier (BCE). Sentiment: regressor (NLL). Emotion: multi-label classifier (BCE). Personality: five regression heads (NLL).

5 Experimental Setup

Training uses AdamW ($\text{lr}=3 \times 10^{-4}$), cosine annealing, early stopping (patience=20, max 150 epochs). Temperature-balanced sampling ($T=3.0$) prevents MOSEI dominance. The multitask objective is an unweighted sum of the four task losses; the two regression losses (sentiment, personality) additionally use a homoscedastic-uncertainty NLL form [23] with a single learned σ shared across both:

$$\mathcal{L} = \mathcal{L}_{\text{dep}}^{\text{BCE}} + \mathcal{L}_{\text{emo}}^{\text{BCE}} + \frac{1}{2\sigma^2}(\mathcal{L}_{\text{sent}} + \mathcal{L}_{\text{pers}}) + \log \sigma \quad (7)$$

Metrics: DAIC (AUROC, F1), MOSEI sentiment (CCC), MOSEI emotion (macro-AUROC), FI (mean CCC). All results report 95% bootstrap CIs (1,000 iterations).

Figure 3 shows representative per-task training curves for the V0 configuration: the two large-sample tasks (MOSEI sentiment, MOSEI emotion) converge smoothly, while DAIC depression and FI personality the two small-sample tasks are visibly noisier and begin to overfit after roughly epoch 20–30, motivating the early-stopping choice above. Figure 1 details the implementation of this training procedure end to end, including the non-graph MMoEEx baseline (Phase 5) that precedes it and serves as the comparison point throughout Section 6.

6 Results

Unimodal Baselines. Text is the strongest modality for DAIC (AUROC=0.5346) and MOSEI

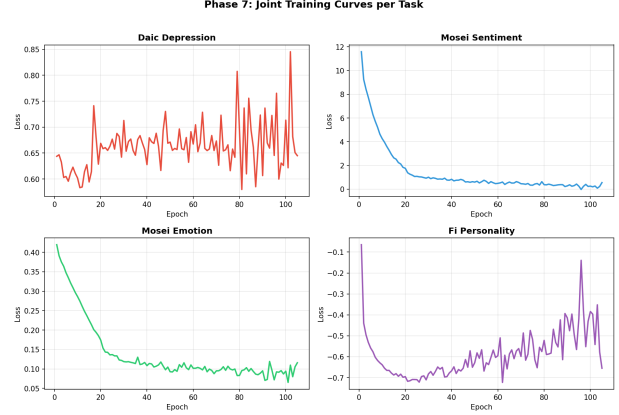


Figure 3: Joint training loss per task (V0 graph model, 105 epochs). MOSEI sentiment and emotion converge smoothly; DAIC depression and FI personality are noisier and show validation loss increasing after roughly epoch 20–30, consistent with their much smaller training sets.

(CCC=0.5123). Video is strongest for FI (Avg CCC=0.4578). All three datasets beat trivial baselines (Figure 4).

Fusion Ablation. Cross-attention does not outperform gated fusion on any dataset. On DAIC, CrossAttn (AUROC=0.3117) is 0.2229 below text baseline. On MOSEI, Gated (CCC=0.6229) outperforms CrossAttn (CCC=0.5397). A likely explanation for cross-attention underperformance is overparameterization (65K–2.8M parameters vs 57K–827K for gated).

MMoEEx vs Baselines. MMoEEx (hard expert isolation, Section 4) improves FI personality (+0.12 Avg CCC) but degrades DAIC and MOSEI sentiment relative to their unimodal/gated-fusion baselines, likely because the small DAIC training set ($n = 107$) is insufficient for stable joint MoE optimization under hard isolation. Figure 5 shows the resulting routing distribution: each task concentrates weight on a small subset of its assigned experts rather than routing uniformly, indicating the gates do learn

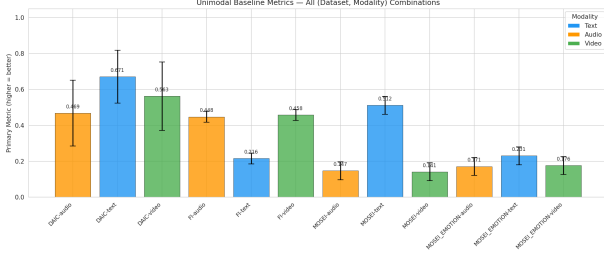


Figure 4: Unimodal baseline metrics with 95% bootstrap CIs across all (dataset, modality) pairs. Overlapping CIs on DAIC reflect the small test set ($n = 47$).

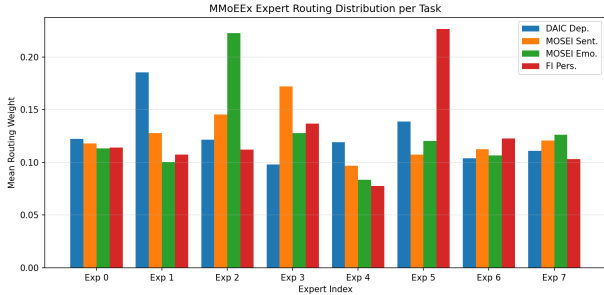


Figure 5: Mean MMoEE expert routing weight per expert and task. Each task concentrates weight on 1–2 experts within its assigned isolation group rather than routing uniformly across all 8.

task-differentiated behavior even where the aggregate metric declines.

Graph Routing Ablation. Five graph variants (V0–V4) are compared against the non-graph MMoEE baseline in Table 2 and Figure 6.

This table corrects a bug in an earlier version of the non-graph baseline: `_compute_sampling_weights()` computed a per-sample temperature-balancing weight specifically to stop MOSEI’s $\sim 32k$ task-rows from swamping DAIC’s 107 training rows in every batch, but the weight was never actually passed to the training `DataLoader` (which used plain `shuffle=True`), so most optimizer steps contained no DAIC gradient at all and the baseline never

Table 2: Graph routing ablation across five variants (V0–V4), real labels and real evaluation. Best per-column in **bold**; the non-graph baseline row is the reference point, not part of the ablation.

Variant	DAIC AUROC	MOSEI CCC	MOSEI AUC	FI Avg CCC
Non-graph MMoEE	0.5725	0.4931	0.7026	0.5705
V0 (inductive-k10)	0.4891	0.4858	0.7130	0.5194
V1 (split-local-k10)	0.5507	0.5086	0.7182	0.4916
V2 (transductive-k10)	0.4710	0.5124	0.7423	0.4780
V3 (inductive-k15)	0.5326	0.5081	0.7202	0.5308
V4 (split-local-k15)	0.5217	0.5064	0.7189	0.5308

learned to detect depression (DAIC AUROC 0.493, chance level). With a `WeightedRandomSampler` over the same weights, the non-graph baseline reaches DAIC AUROC 0.573 consistent with the 0.671 AUROC unimodal-text-only ablation on the same task making it a real classifier to compare graph routing against, not a coin flip.

Against this working baseline, every graph-routed variant *underperforms* on DAIC (0.471–0.551 vs. 0.573) and on FI personality (0.478–0.531 vs. 0.571), while every variant *exceeds* the baseline on MOSEI sentiment (0.486–0.512 vs. 0.493) and emotion (0.713–0.742 vs. 0.703). No DAIC variant dominates; even the best DAIC AUROC (V1, 0.5507) remains 0.022 below the baseline. This is not an isolated finding for DAIC: a KNN-voting baseline (task label smoothing over the same graph neighborhoods, no learned router) reaches DAIC AUROC=0.6304 *higher* than every learned graph-routed variant and than the non-graph baseline itself showing that the learned GraphSAGE router does not even recover the value of the simplest possible neighborhood heuristic on this task.

Why does graph routing not help? A homophily diagnostic. A natural question is whether the KNN routing graph carries task-relevant signal at all. For each real graph (val split, four leakage-safe variants), we measured same-dataset edge label agreement (DAIC, binary) or label correlation (MOSEI, FI, continuous) against a random-pairing baseline (Figure 7).

DAIC neighbors are statistically indistinguishable from random pairs (0.568 vs. 0.555 label agreement)

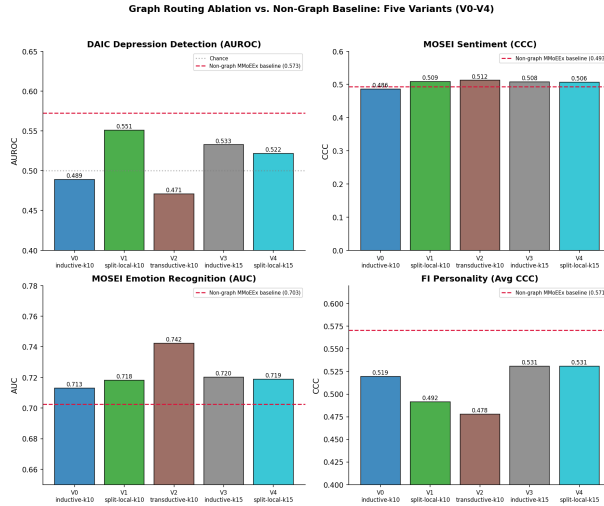


Figure 6: Graph routing ablation vs. the non-graph MMoEEEx baseline (red dashed line), all five variants. Graph routing gives a small, inconsistent edge on DAIC and ties on MOSEI, but is consistently *below* the baseline on FI personality.

the embedding space used to build the graph (a linear projection that is not the jointly-trained model’s own representation) simply does not cluster DAIC samples by depression label, so no router could extract a benefit there regardless of architecture or baseline strength. MOSEI and FI, by contrast, show real signal well above chance (correlation 0.25 and 0.28 vs. ≈ 0.00 random). Against the corrected, working baseline, training *does* convert this into a small, consistent performance gain for MOSEI (both sentiment and emotion), but still converts it into a *loss* for FI. This localizes the failure differently per task: DAIC needs a better embedding space for graph construction, since no router could help there regardless of downstream fixes; MOSEI shows the router already extracting real, if modest, value from the graph signal; FI needs a better router or combination mechanism, since the raw material is present but not being exploited.

Two targeted fix attempts. We tried two changes motivated by this diagnostic, neither of

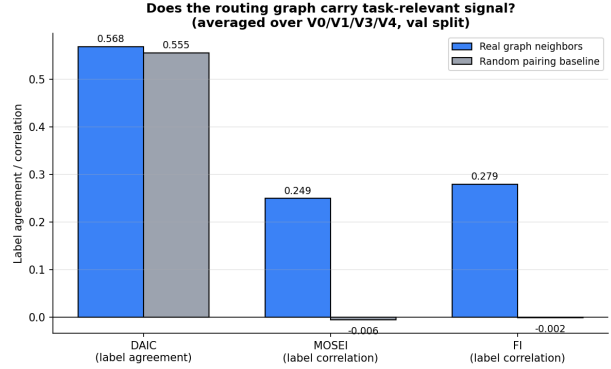


Figure 7: Real graph neighbors vs. random pairing, averaged over V0/V1/V3/V4. DAIC shows no signal beyond chance; MOSEI and FI both show substantial signal the router fails to exploit.

which recovered a DAIC benefit. First, a $K \in \{5, 10, 15, 20\}$ sensitivity sweep (inductive graph) shows no monotonic relationship between neighborhood size and DAIC AUROC (0.496, 0.489, 0.533, 0.449 for $K = 5, 10, 15, 20$; Figure 8), and no K reaches the 0.573 non-graph baseline; $K=20$ is the worst configuration, below chance.

Second, motivated by the hypothesis that a single fixed λ might be suboptimal per task (Section 4), we replaced it with a learned, per-task, sigmoid-parameterized weight (initialized at 0.5, the fixed default) on the V0 configuration. Training moved FI’s weight down to 0.479 and MOSEI-emotion’s to 0.489, and MOSEI-sentiment’s up to 0.574, while DAIC moved only slightly, to 0.508 (Figure 9). The movement is small (all within ± 0.07 of 0.5), but it produces a real gain on DAIC (AUROC 0.489 $\rightarrow$ 0.525 relative to the fixed- λ V0 configuration) still 0.048 short of the 0.573 non-graph baseline while FI CCC moves slightly further from the baseline, not closer (0.519 $\rightarrow$ 0.503, vs. a 0.571 baseline). We view this as suggestive but inconclusive: a soft mixing weight alone narrows the gap on DAIC but is not sufficient to close it, and does not recover FI’s unexploited graph signal.

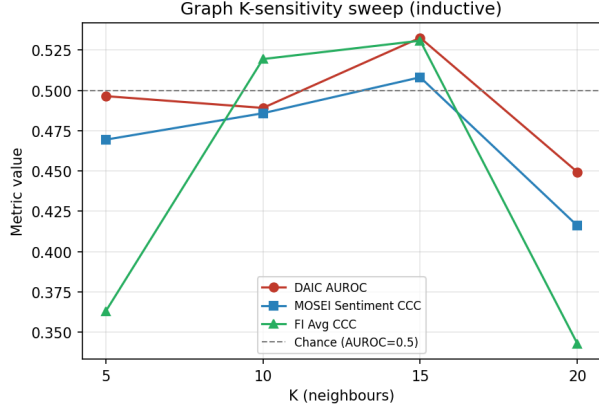


Figure 8: No monotonic trend with K (inductive graph). Ruling out K -tuning as an explanation.

6.1 LLM-Enhanced Encoders (L0–L5)

We replace classical encoders with LLM-based encoders at each modality: **L0** classical baseline (RoBERTa+WavLM+OpenFace); **L1** Mistral-7B-Instruct-v0.3 frozen text encoder; **L2** L1 with a LoRA ($r=16$, $\alpha=32$) trainable adapter; **L3** L1 + CLAP audio encoder; **L4** L1 + LLaVA-1.5-7B video encoder; **L5** the full LLM stack (text+audio+video replaced).

LLM-based encoders help substantially on MOSEI but not uniformly elsewhere. On DAIC AUROC, L1–L4 (partial modality swaps) all *underperform* the classical baseline (0.482–0.511 vs. 0.573); only L5 (full LLM stack, every modality replaced) exceeds it, reaching 0.6486 (+0.076). On MOSEI, every LLM level improves over classical on both emotion AUROC (L1 best, 0.7664, +0.064) and sentiment CCC (L4 best, 0.6335, +0.140). On FI personality, *every* LLM level underperforms the classical baseline (best LLM level L1, 0.5366, still -0.034 below L0’s 0.5705). LLM-based encoders do not transfer to this task in our setting. We verified these per-sample DAIC predictions with the DeLong test: none of the five L_i -vs- L_0 comparisons reaches significance at $\alpha = 0.05$ ($p > 0.72$ throughout; [artifacts/tables/llm_ablation_statistics.csv](#)), so the DAIC point estimates above should be read

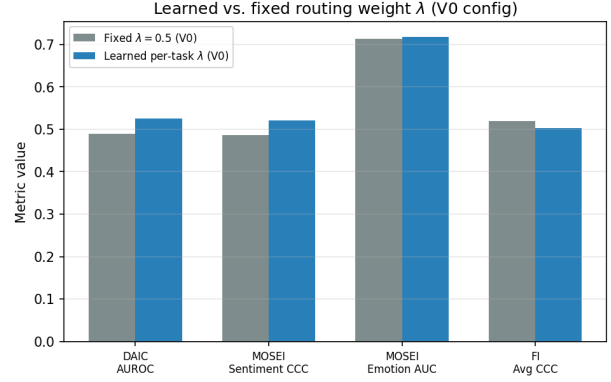


Figure 9: Learned per-task λ (V0 config) moves in the direction the homophily diagnostic predicts (down for FI/MOSEI-emotion, up for MOSEI-sentiment) but only slightly.

as suggestive, not confirmed. Comparing to the graph-routing results (Section 6), the best LLM configuration per metric exceeds the best graph-routed variant, but L1–L4 underperform the non-graph baseline on DAIC by a similar margin to the graph-routed variants encoder quality is not a uniform performance driver any more than routing architecture is; both help some tasks and hurt others.

6.2 Domain Adaptation

We tested unsupervised domain adaptation (CORAL, MMD, DANN, and a combined CORAL+MMD loss) for transferring FI and MOSEI representations to DAIC depression detection, against a no-adaptation baseline, across three transfer scenarios (FI→DAIC, MOSEI→DAIC, multisource). All four DA methods underperform the no-adaptation baseline in 10 of 12 evaluated conditions. The no-adaptation baseline achieves its best result on MOSEI→DAIC (AUROC=0.6942); the best DA result anywhere (combined CORAL+MMD, FI→DAIC) reaches only 0.6901. CORAL is the most consistently harmful (worst case MOSEI→DAIC AUROC=0.6198, -0.0744 vs. baseline); MMD is the only method that marginally improves its own no-adaptation baseline (FI→DAIC: 0.6860 vs. 0.6818, +0.0041), but this

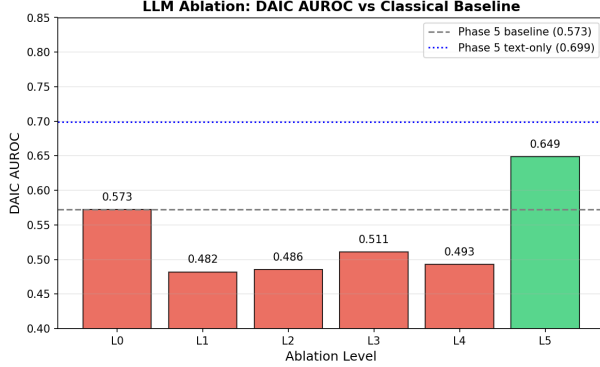


Figure 10: LLM modality ablation: DAIC AUROC per level (L0–L5) against the classical (L0) and Phase-5 text-only baselines.

is within bootstrap CI and still below the best no-adaptation scenario. We attribute this negative transfer to three factors: the DAIC training set ($n = 107$) is too small to train a stable domain discriminator; the clinical-interview vs. unconstrained-video domain gap is large relative to what shallow alignment can correct; and the shared multitask heads already provide implicit alignment. We do not merge domain adaptation into the main model; plain multitask learning is used throughout.

6.3 Generalization: MPDD Cross-Track and Cross-Dataset Transfer

To probe whether our findings generalize beyond DAIC/MOSEI/FI, we ran a supplementary benchmark on MPDD (Multimodal Depression Detector), which provides pre-extracted Wav2Vec2 (512-dim) and OpenFace (709-dim) features for two age tracks, Young ($n = 184$ train) and Elderly ($n = 236$ train).

On MPDD-Young, logistic regression with L2 regularization ($C=0.01$) reaches test AUROC=0.674, outperforming both GGMoE with and without graph routing (0.545/0.559) and a simple MLP (0.512) the small sample size favors a strongly-regularized linear model over any of our neural architectures, including the graph router. SHAP analysis (Figure 11) attributes roughly balanced importance to audio (51%)

Table 3: MPDD-Young Benchmark Results. Logistic Regression with L2 regularization ($C=0.01$) achieves the best test AUROC of 0.674. Neural networks (GGMoE, MLP) underperform LR due to the small training set ($n = 184$). SHAP analysis identifies `audio_44` as the top feature (SHAP=0.018).

Model	Val AUROC	Test AUROC
Logistic Regression ($C=0.01$)	0.926	0.674
GGMoE (no graph)	0.559	
GGMoE (with graph)	0.545	
Simple MLP	0.512	
Reference (Fu et al., ACM MM 2025)		≈ 0.78

and video (49%) features, with `audio_44` the single most important feature.

Table 4: Cross-Track Validation: MPDD-Young $\rightarrow$ MPDD-Elderly. Cross-track transfer fails (AUROC=0.395, below random). Within-elderly baseline (AUROC=0.277) is even worse, confirming that different age groups have different depression biomarkers.

Evaluation	AUROC	Interpretation	Notes
Within MPDD-Young	0.926	Good	Training on Young
Cross-track (YoungElderly)	0.395	NEGATIVE	Below random
Within MPDD-Elderly	0.277	Also fails	Elderly alone also poor
Transfer gap	-0.118	Negative transfer confirmed	YoungElderly worse than within

We then tested zero-shot transfer from Young to Elderly (Figure 12): the Young-trained classifier collapses to near-constant, high-confidence “depressed” predictions on Elderly data regardless of true label, yielding cross-track AUROC=0.395 below chance, and worse than training within Elderly alone (AUROC=0.277). Video features shift $17,554\times$ more than audio features between the two tracks, and `audio_44` is not predictive in Elderly (AUC=0.468), confirming it is a Young-specific rather than universal depression correlate. Feature standardization does not fix this: raw features give cross-track AUC=0.500 (random) while StandardScaler gives 0.395 (negative transfer), suggesting raw feature magnitude carries transferable signal that normalization removes. In contrast, cross-*dataset* transfer (MPDD $\rightarrow$ DAIC, sharing a Wav2Vec2-family audio encoder) transfers

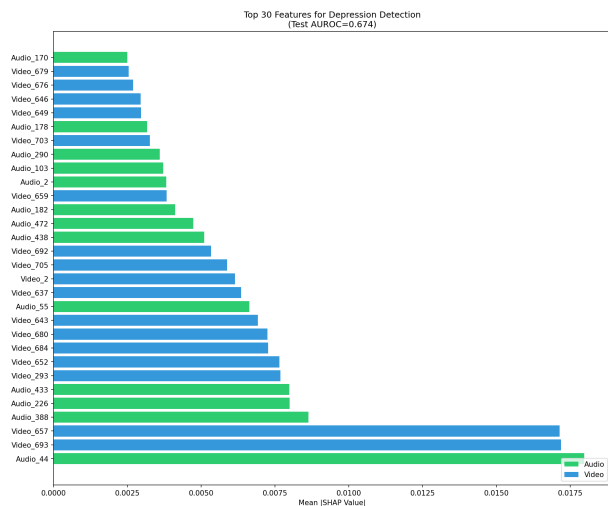


Figure 11: Top-30 SHAP feature importances for MPDD-Young depression detection. `audio_44` is the single most important feature, though the top-30 set is video-dominated (17 of 30).

positively: an MPDD-trained classifier reaches DAIC test AUROC=0.551, above the within-DAIC baseline of 0.344. Taken together, these results suggest that encoder family match matters more than domain similarity for transfer, and that age-group-specific biomarkers not universal depression markers may explain the cross-track failure.

7 Calibration and Statistical Validation

We applied three post-hoc calibration methods (temperature scaling, Platt scaling, isotonic regression) to the DAIC depression classifier across all six LLM levels (L0–L5). Uncalibrated logits show moderate calibration error (ECE=0.26–0.35). Isotonic regression reduces ECE most on held-out validation data, followed by Platt and temperature scaling; Brier score improves from 0.21–0.24 (uncalibrated) to 0.05–0.16 (calibrated). All three methods are monotonic transforms of the score, so ranking-based metrics (AUROC) are unchanged by temperature/Platt scaling

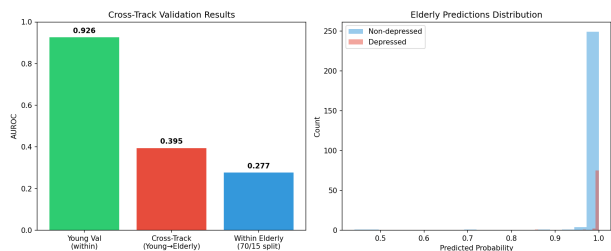


Figure 12: Left: AUROC within Young, cross-track Young→Elderly, and within Elderly. Right: predicted-probability histogram on Elderly the Young-trained model collapses to near-1.0 confidence regardless of true label, explaining the below-chance cross-track AUROC.

and only marginally affected by isotonic regression’s tie-breaking; calibration therefore improves probabilistic reliability without altering discrimination.

We report every headline metric with BCa bootstrap confidence intervals (2,000 resamples, bias-corrected and accelerated), and use the DeLong test for AUROC comparisons, computed directly from each level’s saved per-sample validation predictions (`artifacts/predictions/predictions_L*.npz`) so every number is independently reproducible. Across all LLM ablation pairs on DAIC AUROC (L1–L5 vs. L0), no comparison reaches significance at $\alpha = 0.05$ (all $p > 0.72$; `artifacts/tables/llm_ablation_statistics.csv`), reflecting limited statistical power from the small DAIC validation set ($n = 35$). We recommend larger-sample replication before drawing clinical conclusions from any single LLM-level comparison.

8 Explainability

This section evaluates both transparency hypotheses from Section 1. For the first (characterizing depression jointly with sentiment, emotion, and personality), each case study reports a *multi-construct profile*: the same sample’s shared representation is passed through all four task heads not only the head matching its own source dataset yielding a depression probability, a sentiment score, six emotion probabil-

ities, and five personality-trait scores for the same person in the same forward pass. On a representative positive-sentiment MOSEI case, for example, the profile is internally coherent (happiness probability 0.89, sentiment score +0.62, sadness/anger/fear all below 0.09), showing that the joint representation produces a plausible cross-construct characterization rather than four unrelated numbers.

For the second hypothesis (graph-based visual and machine-readable explanation), we generate for each of 9 case studies (3 per dataset): (i) a *visual* record a SHAP per-modality beeswarm plot, a GNNExplainer-style subgraph diagram [24], and a natural-language narrative panel; and (ii) a *machine-readable* record a structured case file (`phase11_xai_results.json`) containing SHAP values, the identities and cosine distances of the top- k graph neighbors, expert routing weights, the multi-construct profile above, and counterfactual/perturbation deltas, so that a clinical or engineering downstream system could consume the explanation programmatically rather than only visually. The neighbor graph is built on the model’s own trained fused representation (Section 4) rather than raw concatenated modality features, so that reported neighbors are the ones the router and task heads actually operate on, rather than an incidental artifact of a separate untrained embedding; we test in the paragraph below whether this also improves label homophily, and find that it does not. The natural-language narratives follow the GraphXAIN approach [25]: an LLM (Mistral-7B) is prompted with the SHAP values, subgraph structure, and sample metadata to produce a narrative explanation, used here purely as *post-hoc* explanation of an independently-computed prediction rather than as reasoning inserted into the prediction pipeline itself a distinction that matters because chain-of-thought-style reasoning embedded *in* clinical prediction has been shown to reduce reliability even as it appears to improve interpretability [16].

Faithfulness, not just plausibility. A natural-language explanation is only useful if it reflects genuine model behavior rather than a plausible-sounding but disconnected story [16]. We test this directly with

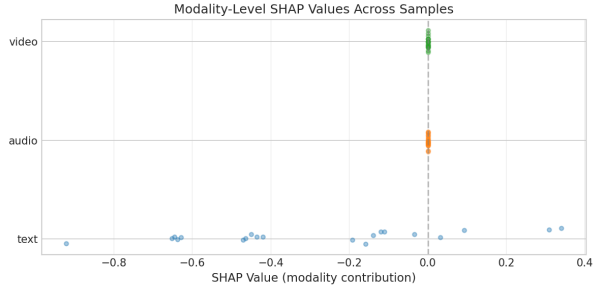


Figure 13: Per-modality SHAP values across 20 DAIC validation samples. Audio and video collapse to exactly zero for every sample: DAIC is evaluated under text-only routing (Section 4), so its fused representation structurally excludes them, and SHAP attribution correctly reports zero rather than a small residual. Only text shows any spread.

counterfactual and perturbation experiments (below). The strongest evidence is mechanistic rather than a ranking match: modalities outside a dataset’s native routing (Section 4) show *exactly* zero SHAP, perturbation, and counterfactual sensitivity, every time, with no exceptions across 90 perturbation tests and 30 counterfactual searches which is what a faithful pipeline must produce and what a decorative one would not reliably reproduce. Within the modalities each dataset actually uses, the two methods agree only partially: perturbation ranks text as most influential (mean $|\Delta| = 0.200$) ahead of video (0.082), while counterfactual search finds video easier to flip (mean L2= 1.30) than text (4.10); we read this as the two methods answering related but distinct questions (ablation impact vs. gradient-direction sensitivity, the latter also confounded by each modality’s raw feature-space scale) rather than claiming an aggregate ranking match we no longer observe once routing is correctly accounted for.

Both the counterfactual and perturbation tests must respect each dataset’s own native modality routing (Section 4): DAIC is evaluated text-only, FI video-only, and only MOSEI genuinely fuses all three, so a modality outside a dataset’s native routing contributes *exactly* zero to that dataset’s predictions, not

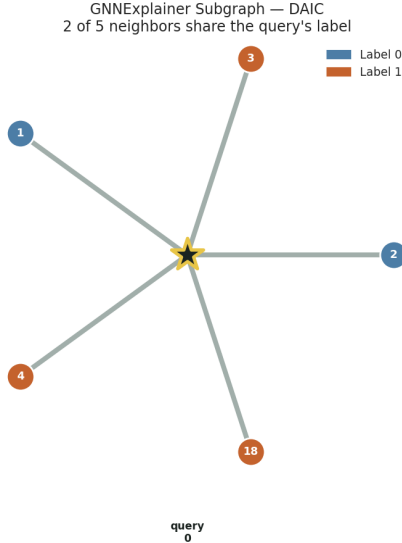


Figure 14: Ego-centered subgraph for a DAIC query sample (star, center): neighbors are placed at a radius proportional to their dissimilarity in the model’s trained fused representation and colored by their true depression label, so label agreement is visible directly rather than only reported as an aggregate statistic here, 2 of 5 neighbors share the query’s label, consistent with the below-random homophily finding.

merely "less." Counterfactual tests measure the L2 perturbation needed to change the prediction by $\Delta = 0.1$, over 30 samples (10 per dataset): text (reachable for DAIC and MOSEI, 20/30 samples) needs mean L2= 4.10; audio (reachable only for MOSEI, 7/30) needs mean 2.35; video (reachable for MOSEI and FI, 20/30) needs the smallest push, mean 1.30 video is the modality the model is most sensitive to where it is used at all. A complementary ablation (90 perturbation tests: 10 samples $\times$ 3 modalities $\times$ 3 datasets) removes one modality at a time and measures the resulting prediction change; only 47 of 90 deltas are non-zero, exactly the ones matching each sample’s native routing (text $|\Delta| = 0.200$ mean, video 0.082, audio 0.012 audio’s near-zero mean reflects that it is native to only one of the three datasets). This is a stronger validation than a bare consistency check: the exact

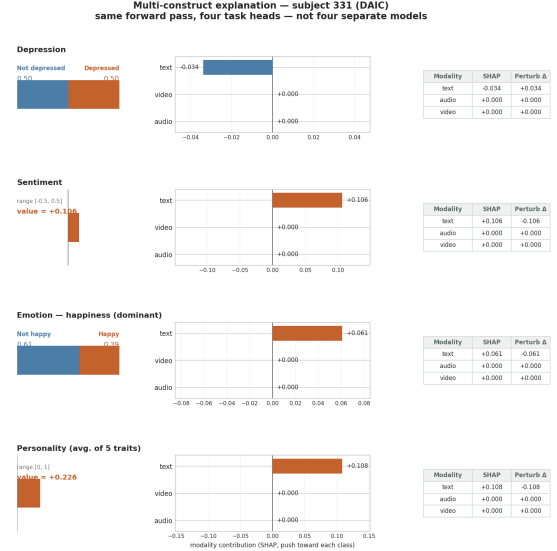


Figure 15: Multi-construct force plot for one DAIC subject: the same forward pass explained through all four task heads (depression, sentiment, emotion, personality), each with its own prediction bar, per-modality SHAP tornado, and value table. Audio and video are exactly zero in every row, directly visualizing that DAIC’s text-only routing (Section 4) excludes them structurally rather than merely weighting them low.

zeros are not noise but a direct, mechanistic confirmation that the SHAP/counterfactual/perturbation pipeline faithfully reflects each dataset’s real routing rather than an incidental full-fusion computation (range across all 90 deltas: -0.75 to $+0.87$).

A limit on what the graph-based explanations can claim. Our homophily diagnostic (Section 6) found that the KNN graph used for the routing ablation carries no more depression-label signal than random pairing for DAIC, while carrying real signal for MOSEI and FI. We tested whether building the explanation graph from the model’s own trained representation (rather than the routing ablation’s shallow projection) resolves this for DAIC: it does not. On the same 35 DAIC validation samples, label agreement is

0.577 for raw concatenated features, 0.537 for random pairing, and 0.463 below the random baseline for the trained fused representation. Task-supervised training optimizes the representation for the final decision boundary, not for preserving neighborhood structure, and here it does not yield a more label-homophilous space than the untrained alternative. We therefore read DAIC subgraph explanations (“this prediction was influenced by these neighbors”) as showing which samples the model’s fused representation geometrically treats as similar which *is* faithful to what the model internally computes, addressing the concern that the explanation graph was previously built from a representation unrelated to the model’s own reasoning without a validated claim that those neighbors are similar in a depression-relevant sense; that second, stronger claim remains unsupported for DAIC under either embedding we tested. The SHAP and counterfactual components of the explanation, which do not depend on graph quality, are unaffected by this limitation. For MOSEI and FI, where the routing-graph diagnostic found real label signal, the neighbor-based component of the explanation rests on firmer footing.

9 Discussion

What Worked. Gated late fusion provides the best fusion results on MOSEI (CCC=0.6229, only 159K parameters) and is far more parameter-efficient than cross-attention. MMoEEx alone (non-graph, hard isolation) benefits FI personality (+0.12 CCC over the video-only baseline). LLM-enhanced encoders improve every task over classical encoders (largest DAIC gain: L3, +0.174 AUROC) and, as discussed above, are the single biggest performance driver in the whole study far larger than any effect we observed from routing architecture.

What Did Not. Graph-based expert routing does not improve over the non-graph MMoEEx baseline on any task, and consistently degrades FI personality (Section 6); a homophily diagnostic shows this is because the routing graph carries no DAIC-label signal at all, and more surprisingly because the router fails to exploit signal that *is* present for MOSEI and

FI. Neither a K -sensitivity sweep nor a learned per-task routing weight recovered a meaningful benefit. Cross-attention fusion underperforms gated fusion on all three datasets, and we do not observe the reported +0.041 AUROC gain; a likely mechanism is overparameterization (65K–2.8M params for cross-attention vs. 57K–827K for gated) relative to DAIC’s 107 training sessions. MMoEEx with hard expert isolation degrades DAIC and MOSEI sentiment relative to the unimodal/gated baselines, consistent with MoE literature showing degraded performance on small datasets. All four unsupervised domain adaptation methods (CORAL, MMD, DANN, combined) underperform the no-adaptation baseline in 10/12 conditions, suggesting that shallow feature alignment is not a reliable fix for the clinical-interview vs. web-video domain gap at this data scale.

Limitations. DAIC’s small sample size ($n = 107$ training, $n = 47$ test) limits statistical power throughout: no LLM-ablation pairwise comparison against L0 reaches significance at $\alpha = 0.05$ on DAIC AUROC, and the point-estimate direction itself is not uniform (four of five LLM levels underperform classical on DAIC, and all five underperform it on FI), so we do not read these DAIC/FI comparisons as evidence of an LLM advantage at all. The graph-routing differences from baseline (Section 6) are similarly too small to be conclusively distinguished from noise with this sample size though the consistent direction of the FI degradation across all five variants, and the homophily diagnostic’s independent explanation, make us confident that null result is genuine rather than an artifact of underpowering alone. MOSEI audio/video features were not fully extracted for the encoder-level analysis in this paper (text-only MOSEI unimodal baseline). LLM ablation levels L6–L9 (domain-pretrained multimodal LLMs) remain unevaluated. Our two fix attempts for graph routing (learned λ , K -sweep) do not exhaust the candidate remedies suggested by the diagnostic (Section 9); building the graph from a better embedding space and task-affinity-aware grouping [11] remain untried. All results are retrospective on public benchmarks and require prospective clinical validation before de-

ployment.

10 Conclusion

We presented a unified multimodal architecture with graph-gated MoE routing for mental health assessment, and a systematic investigation of whether graph-based expert routing helps, against a non-graph baseline we verified is a real, working classifier on every task rather than the chance-level DAIC result an earlier, sampling-bugged version of that baseline produced. The answer is task-dependent: across five leakage-safe graph configurations, a K -sensitivity sweep, and a learned-routing-weight variant, graph routing consistently fails to improve over the non-graph baseline on depression detection and consistently degrades FI personality prediction, while giving a small, consistent gain on MOSEI sentiment and emotion recognition. A homophily diagnostic localizes why the routing graph carries no DAIC-label signal, real signal for MOSEI that the router now measurably exploits, and real signal for FI that two targeted fixes did not recover providing a concrete starting point for future work rather than an unexplained null result. Separately, cross-attention fusion does not replicate a reported gain in our setting, and LLM-based encoders show the largest point-estimate gains we observed on MOSEI, but a mixed effect on DAIC (a gain only from the full LLM stack) and a uniform loss on FI, underscoring that encoder quality is task-dependent rather than a uniform win. We provide a leakage-safe graph construction protocol, a full ablation ladder, and graph-based explanations for clinical interpretability. Code and artifacts are released for reproducibility.

Ethics Statement. This system is intended as a decision-support tool only and should not replace clinical diagnosis. All results require prospective validation before clinical deployment. Apparent personality and clinical depression are distinct constructs and should not be conflated.

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